PRODUCTS OF REGULAR SEQUENCES REMAIN H1-REGULAR AND KOSZUL REGULAR

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot (Xy-pic setup plus theorem declarations and mathematical macros). Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

Editorial note. The citation fields (source_id, source_version, authors, source_url, license.name, source_title) are identical to those of every other card this lane has produced from this chapter, so the batch selector treats them as one source.

Editorial note. The author's proof is terse but complete: it cites the exact input it uses and the deduction from it is a one-step consequence, so this is not a case of an omitted proof. Recorded here because the short length could be mistaken for a hint-only proof by a length-based screen.

Allowed standard prerequisites (editorial).

- The exact sequence comparing the Koszul homologies of the sequences obtained by appending f, fg and g. Lemma \ref{lemma-koszul-mult}, cited at proof lines 7325-7332. Supplies the sequences whose exactness transfers vanishing of the outer groups to the middle group.

Source reference. Original lemma 062G: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7795–7795. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 062C: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7791–7791. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let $f_1, \ldots, f_{r-1} \in R$ be a sequence and $f, g \in R$. Let M be an R-module.*

(1) *If $f_1, \ldots, f_{r-1}, f$ and $f_1, \ldots, f_{r-1}, g$ are M-H_1-regular then $f_1, \ldots, f_{r-1}, fg$ is M-H_1-regular too.*

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(2) *If $f_1, \ldots, f_{r-1}, f$ and $f_1, \ldots, f_{r-1}, g$ are M-Koszul-regular then $f_1, \ldots, f_{r-1}, fg$ is M-Koszul-regular too.*

Proof. By Lemma 062C we have exact sequences

$$H_i(K_\bullet(f_1, \ldots, f_{r-1}, f) \otimes M) \to H_i(K_\bullet(f_1, \ldots, f_{r-1}, fg) \otimes M) \to H_i(K_\bullet(f_1, \ldots, f_{r-1}, g) \otimes M)$$

for all i. □